# Battery-resistor DC circuit

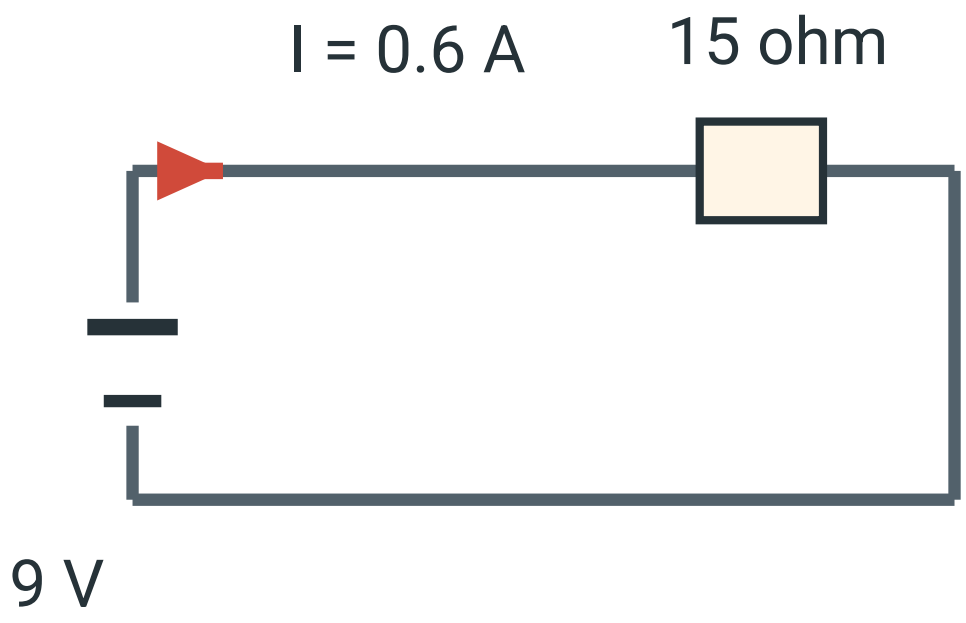
This schematic shows a 9 V source and a 15 ohm resistor. The indicated conventional current is constant at 9 / 15 = 0.6 A. The red arrow moves clockwise along the upper wire to indicate direction; its motion does not represent a changing current magnitude.

```
let supplyVoltage: Number = 9
let resistance: Number = 15
let current: Number = supplyVoltage / resistance
fn indicatorX(t: Number): Number = 58 + 25 * t
diagram batteryResistor {
title: "Battery-resistor DC circuit"
description: "A 9 volt battery drives a 15 ohm resistor; the arrow travels clockwise to indicate conventional current."
width: 320
height: 200
time t from 0 to 4 duration 4
line upperLeftWire {
x1: 50
y1: 130
x2: 188
y2: 130
stroke: "#52616B"
lineWidth: 3
arrow: "none"
}
line upperRightWire {
x1: 218
y1: 130
x2: 250
y2: 130
stroke: "#52616B"
lineWidth: 3
arrow: "none"
}
line leftUpperWire {
x1: 50
y1: 130
x2: 50
y2: 98
stroke: "#52616B"
lineWidth: 3
arrow: "none"
}
line leftLowerWire {
x1: 50
y1: 68
x2: 50
y2: 50
stroke: "#52616B"
lineWidth: 3
arrow: "none"
}
line rightWire {
x1: 250
y1: 130
x2: 250
y2: 50
stroke: "#52616B"
lineWidth: 3
arrow: "none"
}
line lowerWire {
x1: 250
y1: 50
x2: 50
y2: 50
stroke: "#52616B"
lineWidth: 3
arrow: "none"
}
line positivePlate {
x1: 39
y1: 92
x2: 61
y2: 92
```

```
stroke: "#263238"
lineWidth: 4
arrow: "none"
}
line negativePlate {
x1: 43
y1: 74
x2: 57
y2: 74
stroke: "#263238"
lineWidth: 3
arrow: "none"
}
rectangle resistor {
x: 188
y: 118
width: 30
height: 24
stroke: "#263238"
fill: "#FFF5E6"
lineWidth: 2
}
line currentIndicator {
x1: indicatorX(t)
y1: 130
x2: indicatorX(t) + 14
y2: 130
stroke: "#D04A3A"
lineWidth: 4
arrow: "end"
}
text voltageLabel {
x: 20
y: 24
value: "9 V"
fill: "#263238"
}
text resistanceLabel {
x: 180
y: 156
value: "15 ohm"
fill: "#263238"
}
text currentLabel {
x: 88
y: 154
value: "I = ${round(current, 2)} A"
fill: "#263238"
}
}
```

## Battery-resistor DC circuit

A 9 volt battery drives a 15 ohm resistor; the arrow travels clockwise to indicate conventional current.




The horizontal position of the arrow varies with diagram-local time. The current magnitude remains the computed value 0.6 A throughout the displayed interval.